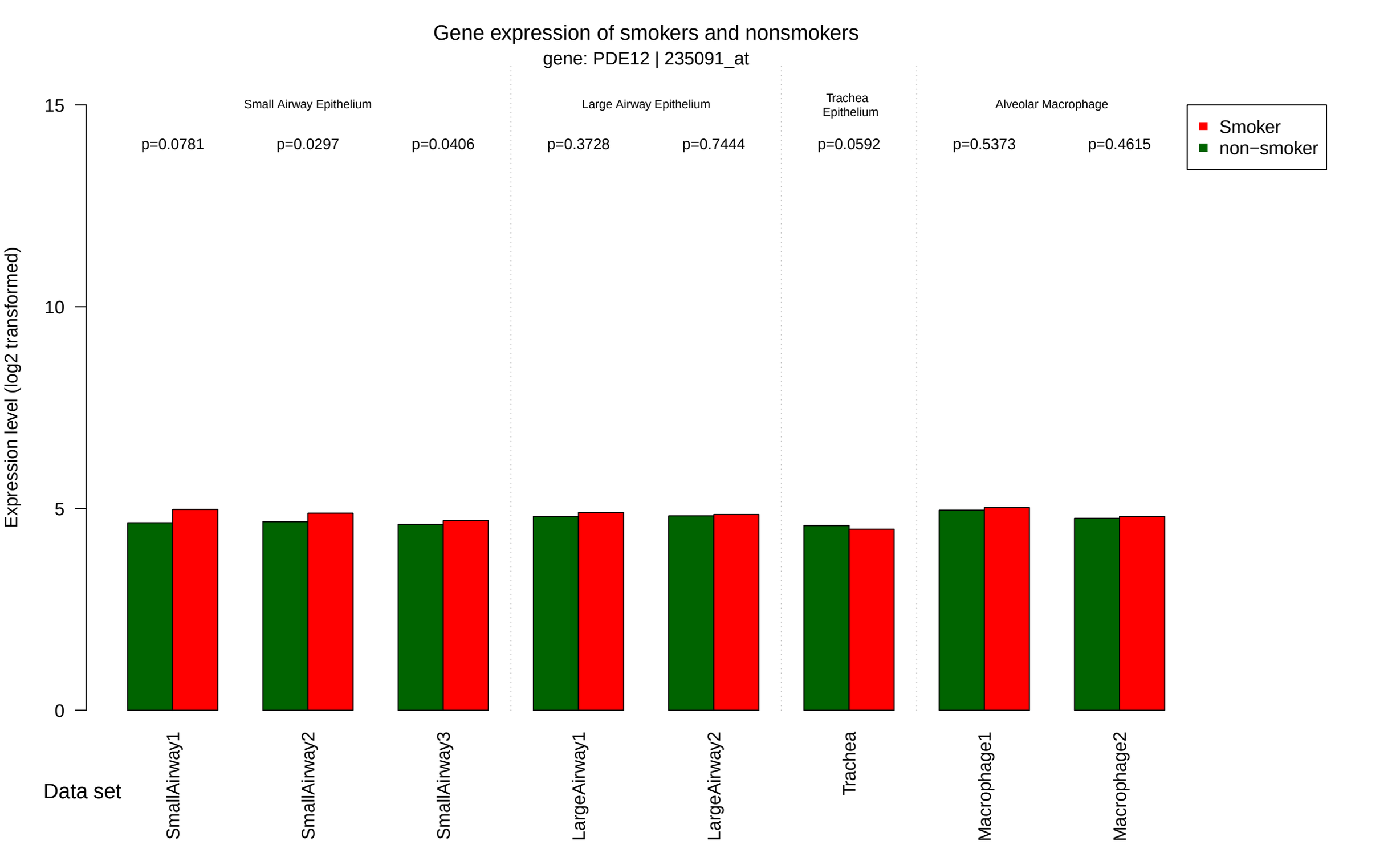
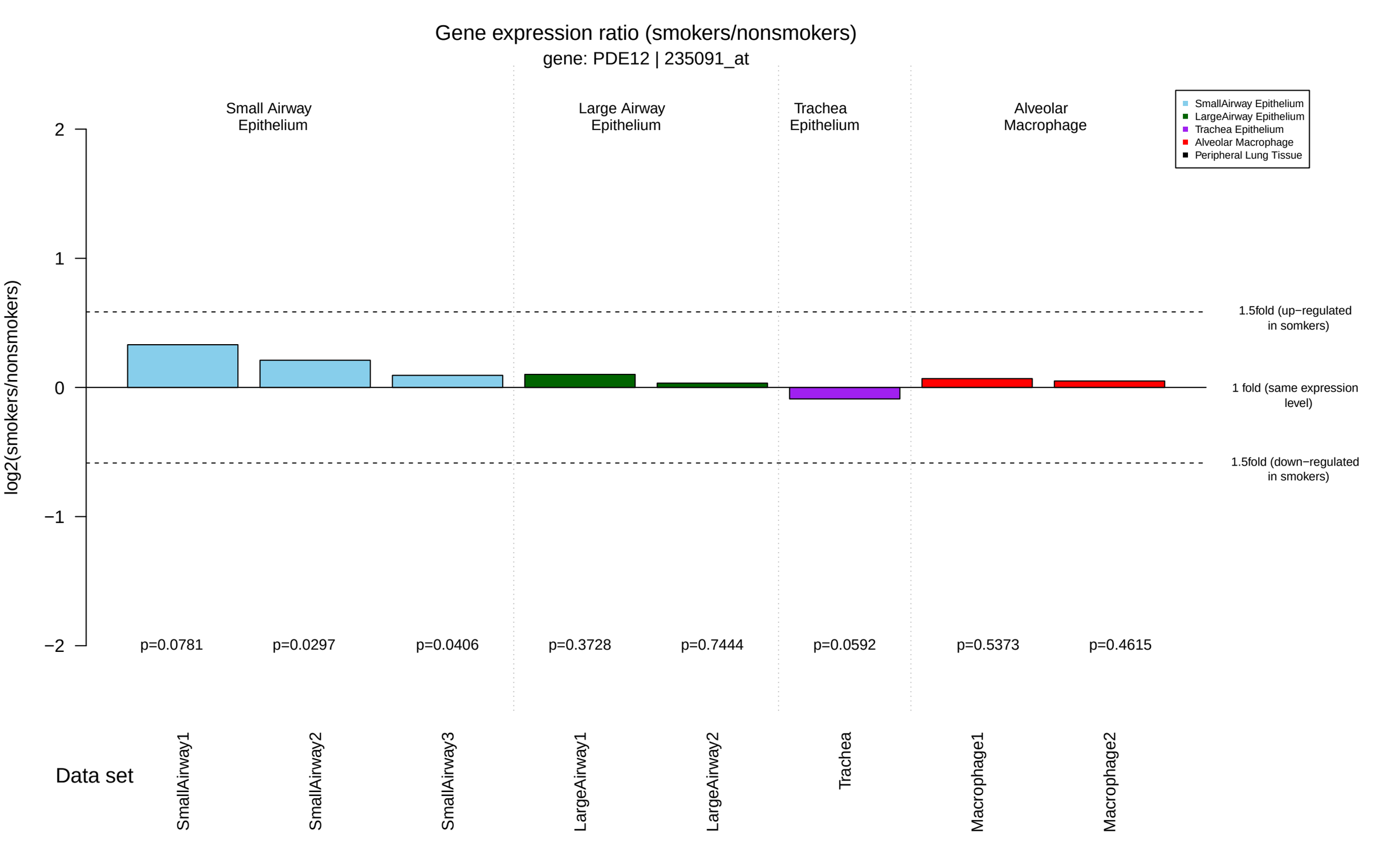
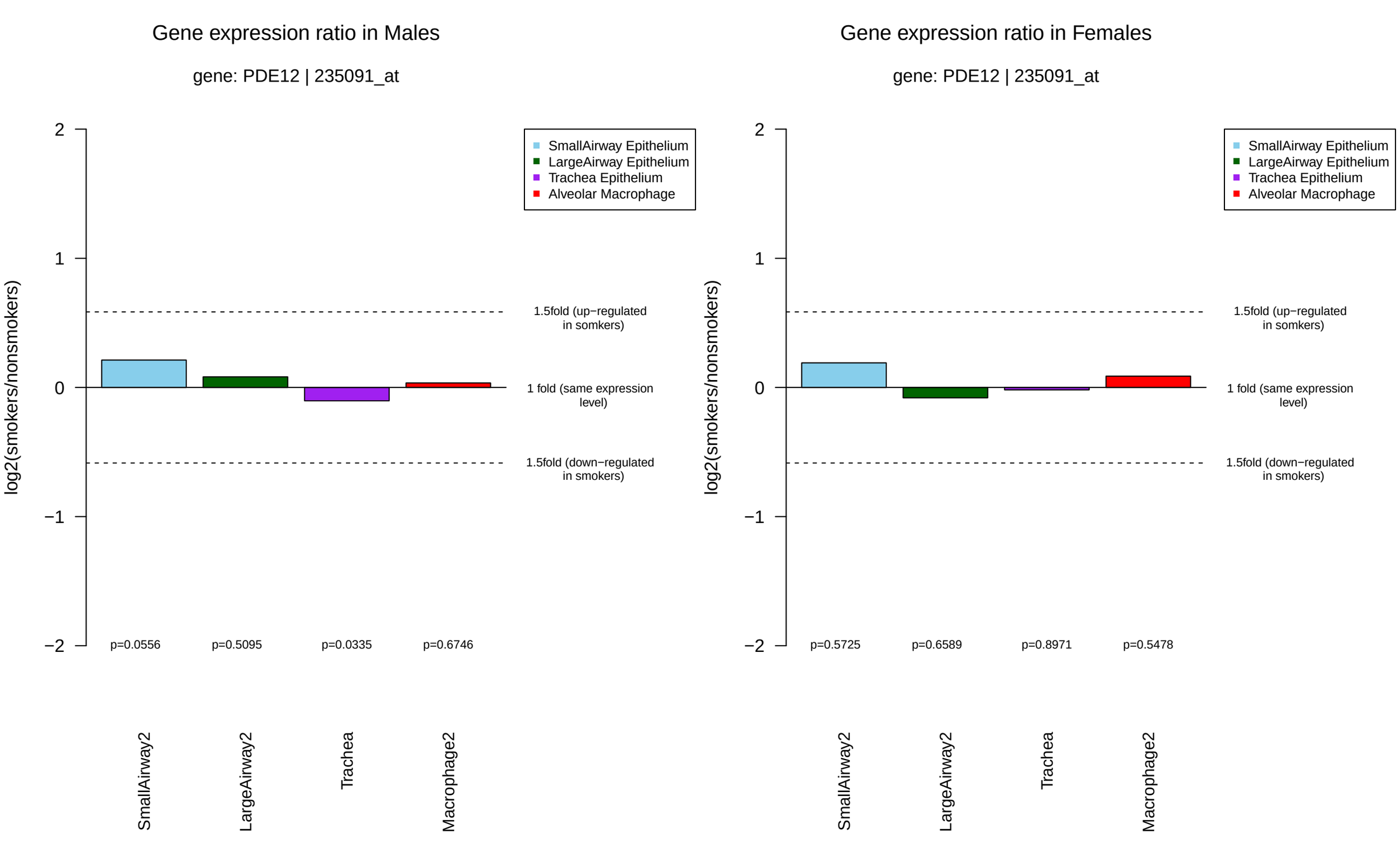
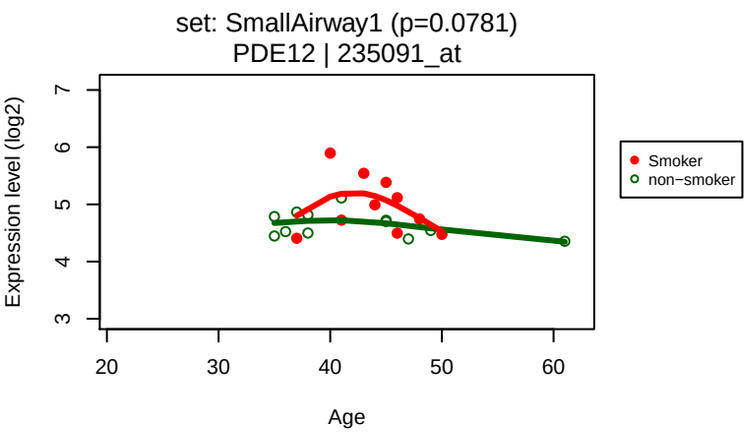
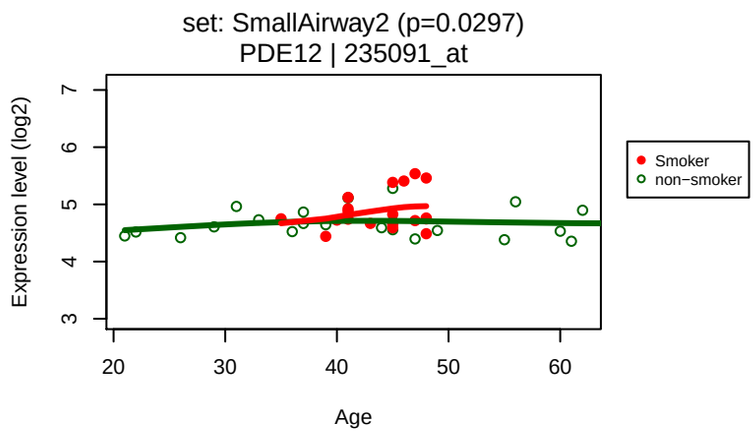
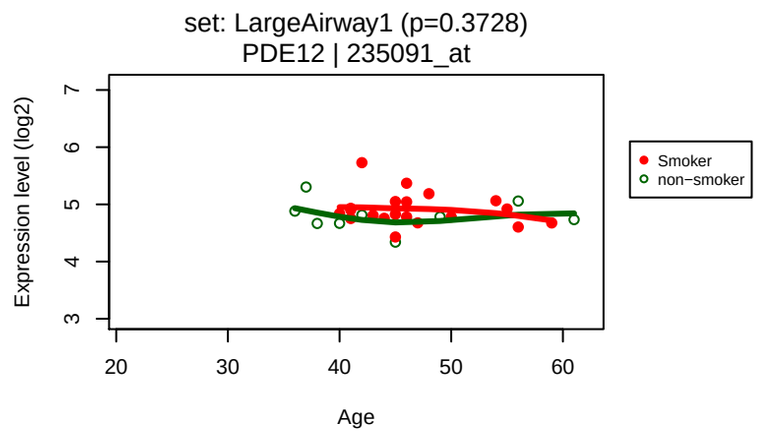

Gene expression of smokers and nonsmokers
gene: PDE12 | 235091_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0781
p=0.0297
p=0.0406
p=0.3728
p=0.7444
p=0.0592
p=0.5373
p=0.4615
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

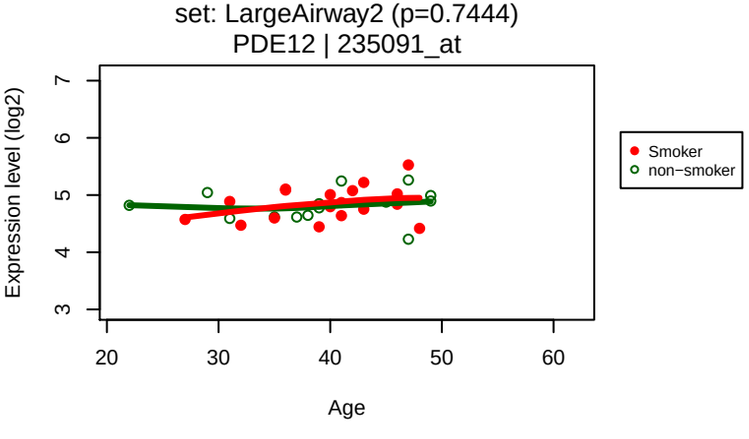

Gene expression ratio (smokers/nonsmokers)
gene: PDE12 | 235091_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0781
p=0.0297
p=0.0406
p=0.3728
p=0.7444
p=0.0592
p=0.5373
p=0.4615
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

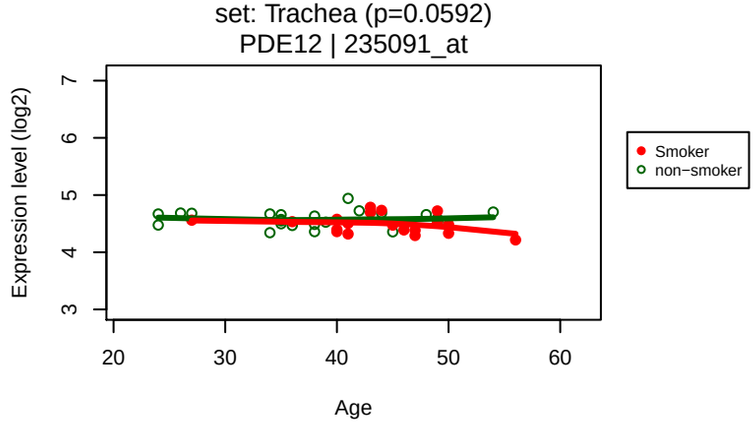

Gene expression ratio in Males
gene: PDE12 | 235091_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0556
p=0.5095
p=0.0335
p=0.6746
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE12 | 235091_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5725
p=0.6589
p=0.8971
p=0.5478
SmallAirway2
LargeAirway2
Trachea
Macrophage2

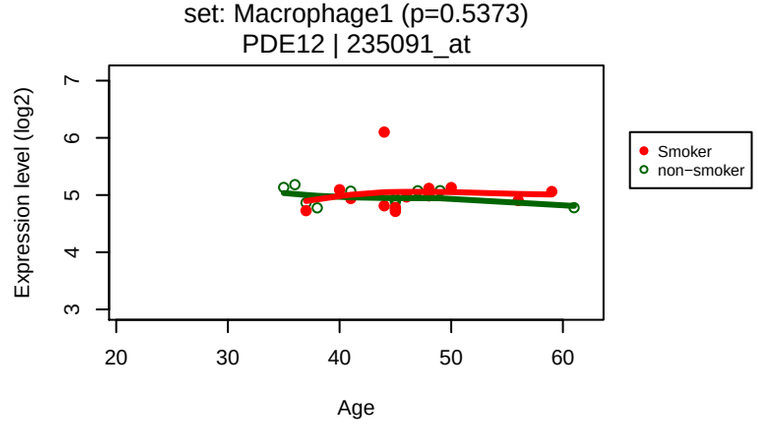

set: SmallAirway1 (p=0.0781)
PDE12 | 235091_at
Expression level (log2)
Age
Smoker
non−smoker

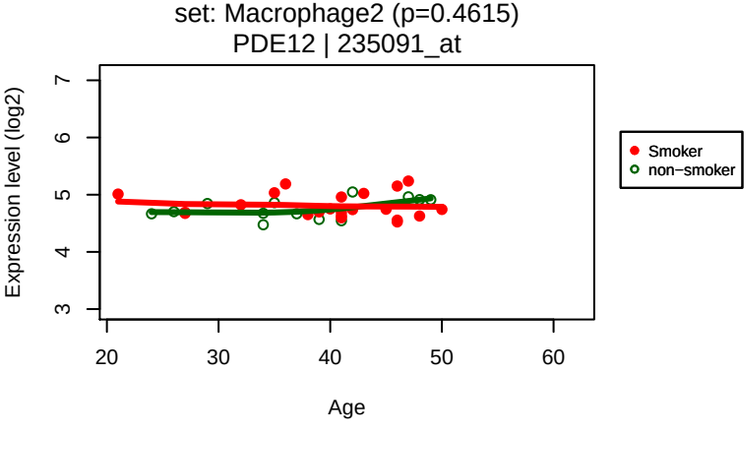

set: SmallAirway2 (p=0.0297)
PDE12 | 235091_at
Expression level (log2)
Age
Smoker
non−smoker

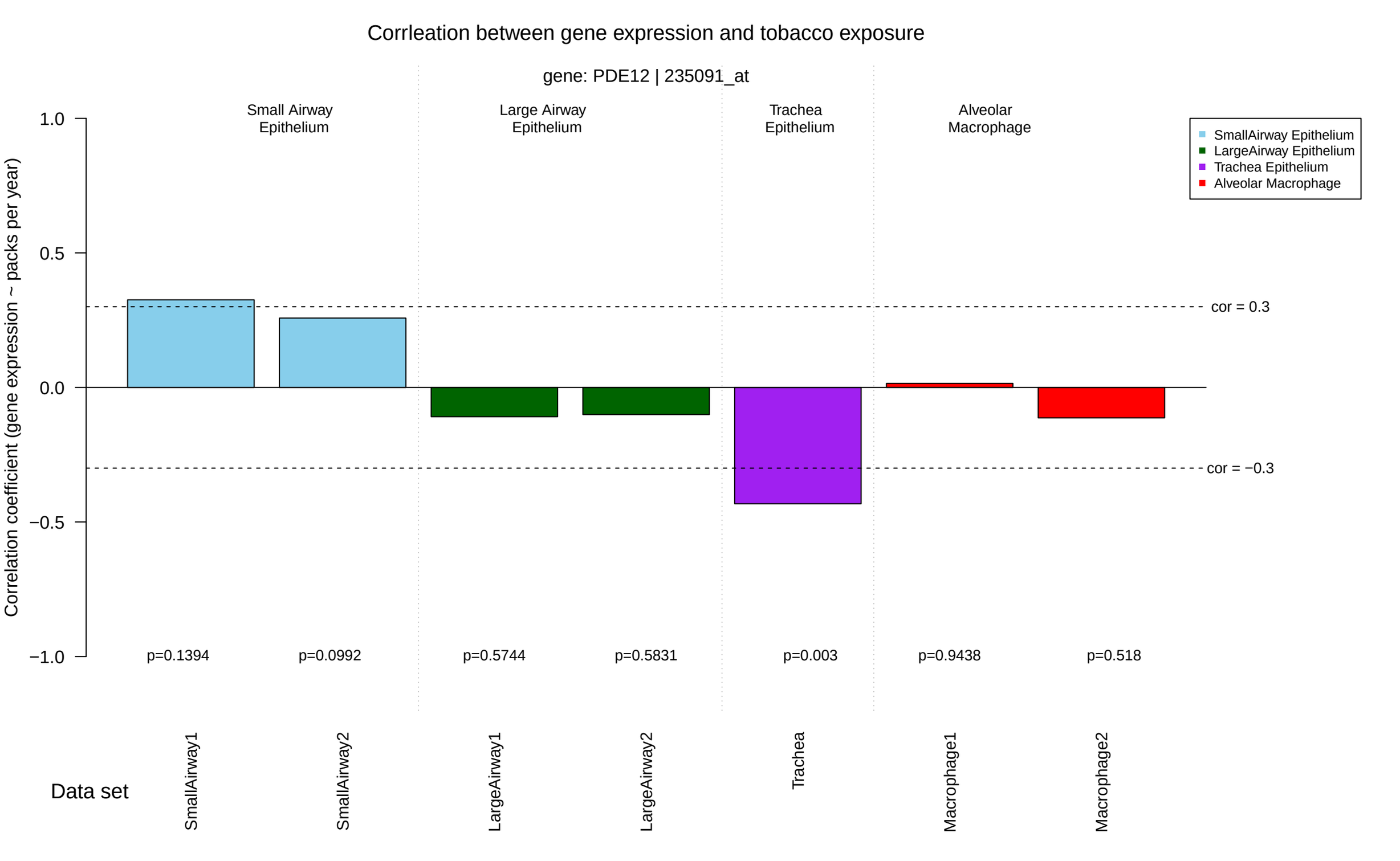

set: LargeAirway1 (p=0.3728)
PDE12 | 235091_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7444)
PDE12 | 235091_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0592)
PDE12 | 235091_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5373)
PDE12 | 235091_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4615)
PDE12 | 235091_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE12 | 235091_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1394
p=0.0992
p=0.5744
p=0.5831
p=0.003
p=0.9438
p=0.518
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2